RELATIVE VOLATILITY RESEARCH

Relative Volatility

Long MRVL / Short SMCI

Trade: Long MRVL / short SMCI straddles
Universe: Equity options Horizon: 5 sessions Status: Conditional

NVDA shocks coincide with more persistent MRVL volatility and greater SMCI normalization. Test whether options price this difference.

SNAPSHOT

Mean ΔRV Gap	**+18.98 pp**
Matched Events	**103**
Price Sample	**2015–2026**
Current IV Gap	**Unverified**
Long Leg	**MRVL straddle**
Short Leg	**SMCI straddle**
Target	**Relative repricing**
Stop	**Loss / time limit**
Status	**Research stage**

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Executive Summary

PROPOSED: Long MRVL / short SMCI straddles **Mean ΔRV difference: +18.98 pp** **103 events**

01
Economic Mechanism

Both companies sell into AI infrastructure. NVIDIA news can alter demand expectations across that supply chain. Hypothesis: MRVL uncertainty persists while SMCI volatility normalizes more quickly.

02
Evidence & Dislocation

Across 103 shocks, MRVL mean RV rises 5.83% and SMCI falls 13.14%. Median changes have the same signs. This identifies a candidate relative trade. Historical IV and net option P&L remain untested.

03
Trade & Failure Conditions

Buy MRVL and sell SMCI ATM straddles only when relative option pricing supports the forecast. Match initial vega and hedge each stock. Company jumps, costs and shifting exposure can invalidate the trade.

Relationship & Economic Rationale

Economic Mechanism

Supermicro supplies NVIDIA-based servers. Marvell supplies custom silicon and data-center connectivity. NVDA news can change shared spending expectations. Different business exposures may produce different volatility responses. The event study tests that hypothesis.

Transmission Channels

SMCI: server demand and deployment timing
MRVL: custom compute and interconnect demand
Hypothesis: different persistence of uncertainty

Why the Link Should Persist

MRVL mean ΔRV exceeds SMCI in 11 of 12 calendar years. The link may reflect common AI exposure, but ordinary volatility mean reversion or company news could explain the pattern.

Key Numbers

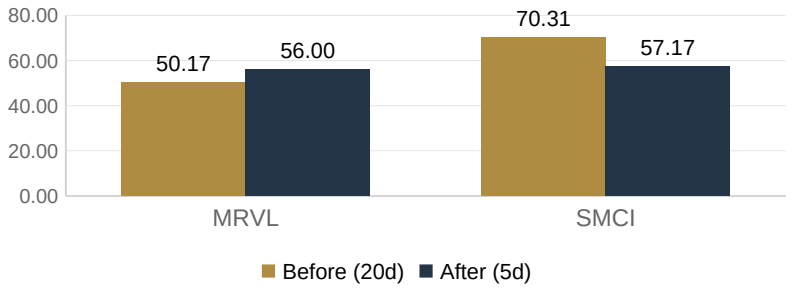
Long Market	**MRVL options**
Short Market	**SMCI options**
Price Sample	**Jan 2015–Oct 2026**
Data Source	**Yahoo / yfinance**
Outcome	**RV change**
Matched Events	**103**
MRVL Mean ΔRV	**+5.83 pp**
SMCI Mean ΔRV	**−13.14 pp**
MRVL Median ΔRV	**+2.64 pp**
SMCI Median ΔRV	**−13.92 pp**
Present IV Gap	**Not established**

Historical Evidence: Different Volatility Responses

How We Measured It

Daily adjusted prices, 2015–2026. NVDA residual versus SPY using 126 prior sessions. Events: |z| ≥ 2. Compare 20-day pre-event SD with next 5-day SD. Day 0 excluded.

Average Annualized Volatility (%)



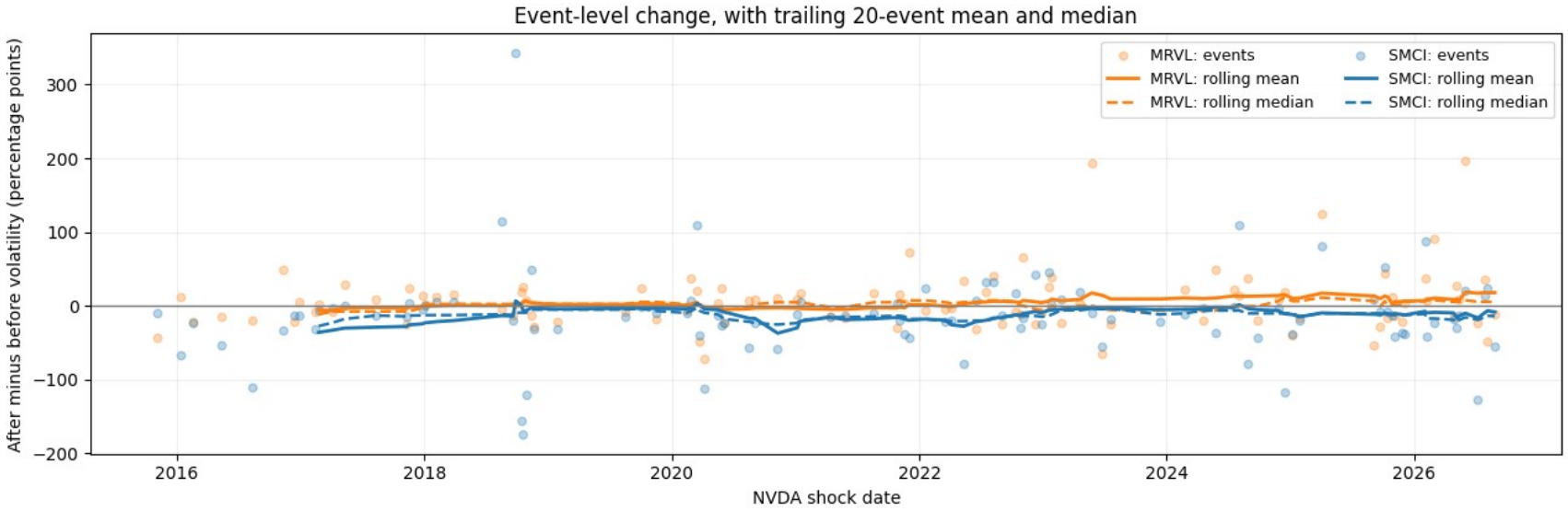
Mean Change by Period (Percentage Points)

Period	MRVL	SMCI	Gap	n	Sign
Full sample	**+5.83**	**−13.14**	**+18.98**	**103**	**M > S**
2023	+17.72	−4.11	+21.83	8	M > S
2024	+14.41	−20.92	+35.33	9	M > S
2025	−0.79	−6.09	+5.30	11	M > S
2026 YTD	+27.50	−15.15	+42.65	11	M > S
2023–2026	+14.49	−11.66	+26.15	39	M > S

MRVL exceeds SMCI on annual mean ΔRV in 11 of 12 years. The 2015 exception has one event. The spread weakens in 2025. These are volatility differences, not P&L.

Event-Level Evidence and Outlier Risk

Rolling means and medians show a changing, uneven relationship



2026 MRVL mean change: +27.50 pp, median: +9.26 pp. Large events influence averages.

Historical Pattern vs. Current Option Pricing

The Relative-Volatility Candidate Requires an Executable Pricing Gap

Metric	Before	After	Change	Interpretation
MRVL mean RV	50.17%	56.00%	+5.83 pp	Average volatility increases after the selected shocks
SMCI mean RV	70.31%	57.17%	−13.14 pp	Higher baseline gives way to lower subsequent volatility
Median ΔRV	MRVL	SMCI	+2.64 / −13.92	Same directional contrast as the mean changes
RV increases	MRVL: 53.4%	SMCI: 29.1%	24.3 pp	Frequency of increasing RV, not the trade win rate
Current IV gap	Not measured	Not measured	Unverified	Current dislocation and option profitability remain open

What Would Establish the Edge?

Economic Hypothesis

Shared AI demand news may leave different persistence of uncertainty. Ordinary mean reversion is an alternative explanation.

Pricing Test

Match maturities and forecast total variance. Compare each leg with its implied pricing, risk premium and execution costs.

Profitability Test

Historical quotes must show positive net delta-hedged P&L. Actual testing was limited by inability to access option pricing data.

Trade Expression

POSITION PARAMETERS

Structure	**Long MRVL / short SMCI vol**
Instruments	**ATM straddles + stock hedges**
Sizing	**Initial vega match + risk cap**
Entry	**Next snapshot, pricing permits**
Target	**Relative pricing normalizes**
Risk/Reward	**Requires option scenarios**
Stop	**Pre-set dollar loss budget**
Horizon	**5 trading sessions**
Exit Plan	**Target, loss limit or day 5**

TRADE RATIONALE

Why This Expression

Long MRVL seeks persistent uncertainty. Short SMCI seeks overpriced volatility normalization. Vega matching reduces exposure to a parallel IV shift.

Catalyst

An observed NVDA shock defines the event. Recheck prices at the next executable snapshot. Same-day closing entry would use unavailable signal information.

What Convergence Looks Like

MRVL IV rises or falls less than SMCI IV, or the gamma-weighted realized-variance advantage exceeds premiums and costs.

Cost of Holding

Include option spreads, theta and daily stock hedges. Add commissions, funding and borrow. Match expiry and use executable exit marks.

Liquidity

Check MRVL and SMCI option spreads, depth and open interest at entry. Same expiry does not guarantee comparable liquidity.

Thesis Invalidation

Exit at the loss limit or day 5. Suspend new entries if forecast benefit or net P&L fails. Vega matching leaves short-gamma and jump risk.

Failure Conditions and Validation Priorities

Risk	Sev	Prob*	Description	Mitigant / Hedge
Structural Break	HIGH	TBD	AI spending changes or customer shifts weaken the economic link. Historical annual differences may not persist.	Monitor rolling responses and forecast errors. Require fresh evidence before scaling the trade.
Selection Bias	HIGH	TBD	Names were selected after screening. A few large events and overlapping windows can inflate confidence.	Reserve new data. Remove largest shocks. Use non-overlapping events and paired block-bootstrap inference.
Hidden Driver	Medium	TBD	Company earnings or sector moves explain the pattern. SMH includes NVDA. Pre-event SMCI volatility may simply mean-revert.	Exclude own earnings. Test ordinary-day controls and an ex-NVDA sector factor. Verify news dates.
Pricing & Costs	Medium	TBD	Options may already price the response. Overnight repricing and trading costs can erase the advantage.	Use actual historical quotes and execution timestamps. Evaluate the full marked portfolio after costs.
Hedge Failure	Medium	TBD	SMCI jumps while MRVL stays quiet. The long straddle in another stock does not cap short-call losses.	Cap short gamma and jump exposure. Stress each leg. Evaluate defined-risk protection separately.